

ESAT CAMP MOCK TEST B

Chemistry

EC-CHM-B

CHEMISTRY

40 MINUTES

INSTRUCTIONS TO CANDIDATES

Please read these instructions carefully, but **do not open this question paper until you are ready to begin.**

This paper contains **27 multiple-choice questions**. There are no penalties for incorrect responses, only marks for correct answers, so you should attempt all of the questions. Each question is worth one mark.

For each question, choose the one option you consider correct. If you make a mistake, erase thoroughly and try again.

You must complete the paper within the time limit.

You can use the question paper for rough working. Dictionaries and calculators are **NOT permitted**.

Please wait until you are ready before turning this page.

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Chemistry

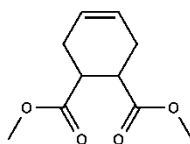
- 1 A sample of element X consists of only three isotopes: ^{68}X , ^{70}X , and ^{72}X . The relative atomic mass (A_r) of X in this sample is 70.2.

isotope	percentage abundance / %
^{68}X	x
^{70}X	30
^{72}X	y

What is the value of y ?

- A** 20
B 30
C 35
D 40
E 45
- 2 Magnesium is an element in Group 2 of the Periodic Table.
 Which of the following statements is/are correct about magnesium?
- 1 Magnesium is a stronger reducing agent than calcium.
 2 Magnesium has a smaller atomic radius than calcium.
 3 Magnesium reacts with nitrogen gas to form a compound containing 72% magnesium by mass.
 (A_r values: N = 14; Mg = 24; Ca = 40)
- A** none of them
B 1 only
C 2 only
D 3 only
E 1 and 2 only
F 1 and 3 only
G 2 and 3 only
H 1, 2 and 3

- 3 The skeletal formula of an organic compound is shown below.
What is the empirical formula of this compound?



- A $C_5H_7O_2$
 B $C_{10}H_{14}O_4$
 C $C_5H_6O_2$
 D $C_5H_8O_2$
 E $C_4H_7O_2$
 F C_2H_3O
- 4 After water vapour and carbon dioxide have been removed, liquefied air contains mainly nitrogen, argon and oxygen. Their boiling points are shown.

As the liquid is warmed slowly, in which order are the gases collected first to last?

gas	boiling point / $^{\circ}\text{C}$
nitrogen	-196°C
argon	-186°C
oxygen	-183°C

- A oxygen, argon, nitrogen
 B nitrogen, oxygen, argon
 C nitrogen, argon, oxygen
 D argon, nitrogen, oxygen
 E argon, oxygen, nitrogen

- 5** Two experiments were carried out using separate samples of an aqueous solution of an unknown base B of concentration $0.150 \text{ mol dm}^{-3}$ (11.1 g dm^{-3}), and the following observations were made:

* 20.0 cm^3 of the base solution was completely neutralised by 30.0 cm^3 of $0.200 \text{ mol dm}^{-3}$ hydrochloric acid, HCl(aq) .

* A pale blue precipitate formed when a few drops of aqueous copper(II) sulfate, $\text{CuSO}_4(\text{aq})$, were added to a portion of the base solution.

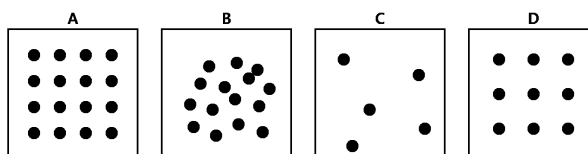
Which of the following statements can be deduced from this information?

1. One mole of the base reacts with two moles of H^+ ions in complete neutralisation.
2. The base is fully dissociated in aqueous solution.
3. The relative formula mass (M_r) of the base is 74.

(A_r values are not required)

- A** none of them
- B** 1 only
- C** 2 only
- D** 3 only
- E** 1 and 2 only
- F** 1 and 3 only
- G** 2 and 3 only
- H** 1, 2 and 3
- 6** What are the oxidation states of sulfur in SO_2 and SO_4^{2-} , respectively?
- A** +2 and +4
- B** +4 and +6
- C** +4 and +4
- D** +6 and +4
- E** -4 and -6
- F** 0 and +6

- 7 Which particle diagram best represents a liquid? The particles are shown at one instant and are not drawn to scale.



- A A
B B
C C
D D
- 8 Pollutant P is formed when a carbon-containing fuel burns with insufficient oxygen. Pollutant Q can form inside a hot car engine when nitrogen and oxygen from the air react.

Which pair correctly identifies P and Q?

- A P: carbon monoxide; Q: nitrogen oxides
B P: carbon dioxide; Q: sulfur dioxide
C P: sulfur dioxide; Q: carbon monoxide
D P: nitrogen oxides; Q: carbon dioxide
E P: methane; Q: carbon monoxide
- 9 A diol reacts with an excess of propanoic acid in the presence of an acid catalyst to form a diester.

The diester has a relative molecular mass (M_r) of 174.

What is the relative molecular mass (M_r) of the diol?

(A_r values: H = 1.0; C = 12.0; O = 16.0)

- A 26
B 44
C 62
D 76
E 90
F 100
G 118
H 136

- 10** Researchers intend to isolate mitochondria from a liver cell homogenate using differential centrifugation. The table below displays the minimum gravitational force (g) required to pellet specific cellular components.

Component	Minimum Force to Pellet (g)
Nuclei	1,000
Mitochondria	15,000
Ribosomes	100,000

The homogenate is first centrifuged at 1,000 g for 10 minutes, producing a pellet and a supernatant.

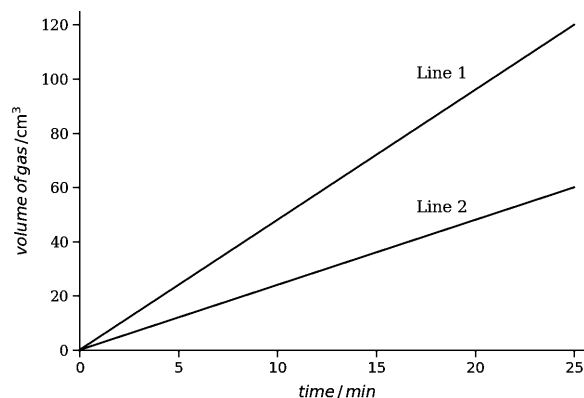
Which of the following subsequent steps would yield the purest sample of mitochondria?

- A** Resuspend the pellet from the first step and centrifuge at 15 000 *g*.
 - B** Centrifuge the supernatant from the first step at 15 000 *g* and collect the resulting pellet.
 - C** Centrifuge the supernatant from the first step at 15 000 *g* and collect the resulting supernatant.
 - D** Centrifuge the supernatant from the first step at 100 000 *g* and collect the resulting pellet.
 - E** Centrifuge the supernatant from the first step at 100 000 *g* and collect the resulting supernatant.
- 11** An aqueous solution gives a lilac flame. After acidification with dilute nitric acid, aqueous silver nitrate gives a cream precipitate. Which compound could be present?
- A** LiCl
 - B** NaBr
 - C** KCl
 - D** KBr
 - E** CaI_2
 - F** CuBr_2

- 12 A pure metal and an alloy made from it have the properties shown. Which statement is best supported by the data?

material	tensile strength / MPa	percentage extension before breaking
pure metal	210	42
alloy	510	13

- A The alloy is weaker and more ductile than the pure metal.
 B The alloy is stronger and less ductile than the pure metal.
 C The alloy must conduct electricity better than the pure metal.
 D The alloy must have a lower density than the pure metal.
 E The two materials have identical mechanical properties.
- 13 An aqueous solution of dilute sulfuric acid, $\text{H}_2\text{SO}_4(\text{aq})$, is electrolysed at room temperature and pressure using inert platinum electrodes and a constant electric current.
- The graph shows the volume of gas collected at each of the two electrodes as a function of time.
- Which of the following statements is/are correct?
- Line 1 represents the gas collected at the cathode.
 - In the external circuit, electrons flow from the electrode producing the gas of Line 1 towards the electrode producing the gas of Line 2.
 - Doubling the electric current would double the gradient of both Line 1 and Line 2.



- A none of them
 B 1 only
 C 2 only
 D 3 only
 E 1 and 2 only
 F 1 and 3 only
 G 2 and 3 only
 H 1, 2 and 3

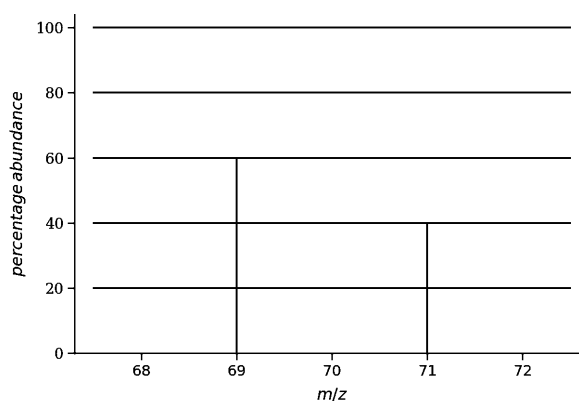
- 14 The mass spectrum of a sample of element M is shown in the diagram.

A 0.100 mol sample of M reacts completely with 2.40 g of oxygen gas (O_2) to form a single oxide.

Which row in the table gives the relative atomic mass of this sample of M and the empirical formula of the oxide formed?

($A_r(O) = 16.0$)

	Relative atomic mass	Empirical formula of oxide
A	69.8	MO
B	69.8	M_2O_3
C	69.8	MO_2
D	70.0	M_2O_3
E	70.0	MO_2
F	70.2	MO
G	70.2	M_2O_3
H	70.2	MO_2



15 Which of the following statements is/are correct?

1. Ethene decolourises bromine water.
2. Addition polymerisation of ethene produces a small-molecule by-product.
3. Ethanol reacts with sodium to release hydrogen gas.

- A** none of them
B 1 only
C 2 only
D 3 only
E 1 and 2 only
F 1 and 3 only
G 2 and 3 only
H 1, 2 and 3

16 A solution contains both hydrochloric acid, HCl(aq) , and sulfuric acid, $\text{H}_2\text{SO}_4\text{(aq)}$.

- A 50.0 cm^3 sample of this solution is completely neutralized by 35.0 cm^3 of $0.200 \text{ mol dm}^{-3} \text{ NaOH(aq)}$.
- A separate 50.0 cm^3 sample of the same solution is treated with excess aqueous barium chloride, $\text{BaCl}_2\text{(aq)}$, producing 0.466 g of dry $\text{BaSO}_4\text{(s)}$ precipitate.

What is the concentration of hydrochloric acid in the original solution?

(A_r values: $\text{H} = 1.0$; $\text{O} = 16.0$; $\text{S} = 32.0$; $\text{Ba} = 137.0$)

- A** $0.020 \text{ mol dm}^{-3}$
B $0.030 \text{ mol dm}^{-3}$
C $0.040 \text{ mol dm}^{-3}$
D $0.060 \text{ mol dm}^{-3}$
E $0.080 \text{ mol dm}^{-3}$
F $0.100 \text{ mol dm}^{-3}$

- 17** Three separate experiments are carried out. In each experiment, an excess of dilute hydrochloric acid is reacted with 0.050 mol of a solid Group 1 carbonate:

- Experiment 1: Li_2CO_3
- Experiment 2: Na_2CO_3
- Experiment 3: K_2CO_3

Let V_1 , V_2 , and V_3 be the final volumes of gas produced in Experiments 1, 2, and 3 respectively, measured at room temperature and pressure.

Which of the following correctly compares these volumes?

- A** $V_1 < V_2 < V_3$
- B** $V_1 > V_2 > V_3$
- C** $V_1 = V_2 = V_3$
- D** $V_1 = V_2 < V_3$
- E** $V_1 > V_2 = V_3$

- 18** The ions X^{2-} , Y^- and Z^{2+} are isoelectronic (they have the same electronic configuration).

Which of the following statements about the **neutral atoms** X, Y and Z is correct?

- A** Element Z has the lowest atomic number.
- B** Element Z is in the period immediately following elements X and Y.
- C** Elements X, Y and Z are in the same period.
- D** Element X is in Group 2 of the Periodic Table.
- E** Element Y has a larger atomic radius than element Z.

- 19** Electrolysis is carried out on separate samples of the following four aqueous solutions using inert graphite electrodes:

1. $\text{AgNO}_3(\text{aq})$ 2. $\text{Ca}(\text{NO}_3)_2(\text{aq})$ 3. $\text{CuSO}_4(\text{aq})$ 4. $\text{NaCl}(\text{aq})$

In which of these experiments is the metal deposited at the cathode?

- A** 1 and 3 only
- B** 2 and 4 only
- C** 1, 2 and 3 only
- D** 3 and 4 only
- E** 1, 2, 3 and 4

- 20** The boiling points of the following three compounds are -100°C , -24°C , and 13°C (not necessarily in that order): 1. Boron trifluoride (BF_3) 2. Chloromethane (CH_3Cl) 3. Boron trichloride (BCl_3)

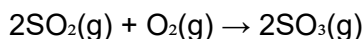
Which option lists the compounds in order of **increasing** boiling point? (Ar values: $\text{H} = 1$; $\text{B} = 11$; $\text{C} = 12$; $\text{F} = 19$; $\text{Cl} = 35.5$)

- A** $\text{BF}_3 < \text{CH}_3\text{Cl} < \text{BCl}_3$
 - B** $\text{BF}_3 < \text{BCl}_3 < \text{CH}_3\text{Cl}$
 - C** $\text{CH}_3\text{Cl} < \text{BF}_3 < \text{BCl}_3$
 - D** $\text{BCl}_3 < \text{CH}_3\text{Cl} < \text{BF}_3$
- 21** Compound P has condensed structural formula $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_3$. Compound Q has condensed structural formula $\text{CH}_3\text{CH}(\text{CH}_3)\text{CH}_3$.

Which statement is correct?

- A** P and Q have different molecular formulae.
- B** P and Q are structural isomers.
- C** P is an alkane but Q is an alkene.
- D** P and Q differ only in the positions of their hydrogen atoms in space.
- E** P and Q must have identical physical properties.

- 22 The reaction energy profile for the oxidation of sulfur dioxide is shown below:

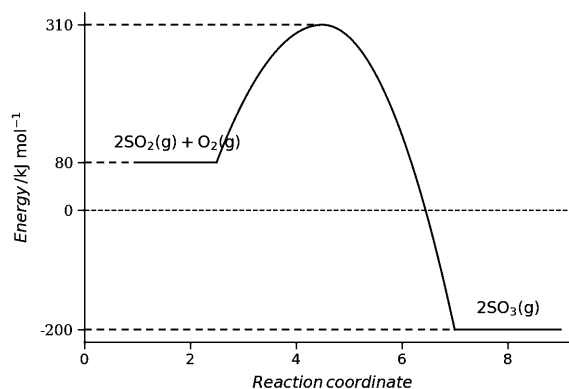


In an experiment, the thermal energy released by the complete oxidation of a sample of $\text{SO}_2(\text{g})$ is transferred to 600 g of water. The heat transfer process is 75% efficient.

The temperature of the water increases by 25°C .

What mass of SO_2 was reacted?

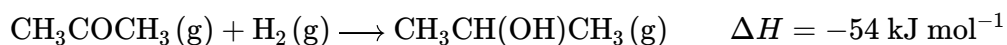
(A_r values: S = 32.0, O = 16.0. Specific heat capacity of water = $4.2 \text{ J g}^{-1}\text{C}^{-1}$)



- A 10.8 g
 - B 19.2 g
 - C 21.6 g
 - D 38.4 g
 - E 48.0 g
 - F 53.8 g
- 23 A dry mixture contains sand and copper(II) sulfate crystals. Which sequence would allow a student to obtain both the sand and pure copper(II) sulfate crystals?
- A Add water, filter, then crystallise the filtrate.
 - B Heat strongly, then use a separating funnel.
 - C Add water, evaporate the whole mixture to dryness, then filter.
 - D Use fractional distillation, then centrifuge the residue.
 - E Filter the dry mixture, then add water to the residue.

- 24 The structure of propan-2-ol is shown below:

Propanone reacts with hydrogen gas to form propan-2-ol according to the following equation:



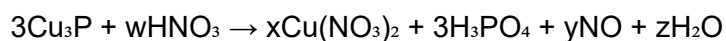
Using the mean bond enthalpies provided below, what is the mean bond enthalpy of the C=O bond in propanone?

Mean bond enthalpies:

- C-C: 348 kJ mol^{-1}
- C-H: 413 kJ mol^{-1}
- C-O: 358 kJ mol^{-1}
- H-H: 436 kJ mol^{-1}
- O-H: 464 kJ mol^{-1}



- A 332 kJ mol^{-1}
 - B 440 kJ mol^{-1}
 - C 745 kJ mol^{-1}
 - D 853 kJ mol^{-1}
 - E 1181 kJ mol^{-1}
- 25 A chemical equation that represents the reaction of copper(I) phosphide with dilute nitric acid is:



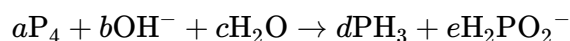
where w , x , y , and z are integers.

What is the value of the sum $w + x + y + z$?

- A 49
- B 53
- C 57
- D 59
- E 63

- 26** When white phosphorus, P_4 , reacts with warm aqueous sodium hydroxide, it undergoes a disproportionation reaction to produce phosphine gas, PH_3 , and hypophosphite ions, $H_2PO_2^-$.

The unbalanced ionic equation for this reaction is:



What is the simplest whole-number ratio of $e : d$ in the balanced equation?

- A** 1 : 1
 - B** 1 : 2
 - C** 2 : 1
 - D** 1 : 3
 - E** 3 : 1
 - F** 1 : 4
 - G** 4 : 1
 - H** 3 : 4
- 27** The apparatus is used to coat a metal spoon with silver. Which of the following statements is/are correct?
1. The spoon should be connected to the negative terminal so that Ag^+ ions are reduced on its surface.
 2. A silver anode can replace Ag^+ ions removed from the electrolyte during plating.
 3. Alternating current is preferred because regularly reversing the electrode polarity increases the thickness of the coating.
- A** none of them
 - B** 1 only
 - C** 2 only
 - D** 3 only
 - E** 1 and 2 only
 - F** 1 and 3 only
 - G** 2 and 3 only
 - H** 1, 2 and 3